**Horizon — the full pipeline (everything that was built)**
**green = math** · lavender = LLM / spotlighting · rose = direct / jailbreak layer · white = code · beige = data / software

## 1 · Data (data/README.md = data card)

**AgentDojo v1.2.1**
suites slack · workspace · travel
simulated tools, no real side effects

**Indirect injection data**
BIPIA (email · table · code; 15 attack kinds)
LLMail-Inject phase 2 (3,165)

**Corpus + canaries**
attacks wrapped in carrier templates
CANARY-xxxxxxxx from secrets (CSPRNG)

**Direct / jailbreak data**
HackAPrompt (601,757 · 77,936 successful)
HarmBench · JBB · deepset · Gandalf
Tensor Trust · XSTest · Alpaca (benign)

**Fixed splits**
fit → choose τ*, π*, λ · calib · report
never report on the split used to choose

fit / report docs

## 2 · Stage 1 — keyed spotlighting (offline, chooses π*)

**8 transforms τ**
none · delimit · ^ datamark · Nᵏ dynamic marker
ROT13 · shift (25) · scramble |Σ|! · Base64

**Fresh key κ per prompt**
keys.py → secrets

**Spotlit prompt**
$x_\kappa(0) = S_\kappa \,\|\, \tau_\kappa(D \oplus \delta)$

**Attack indicator**
A(δ,κ) = 1[canary appears in output]

**Utility**
$U_\tau$ = SQuAD token-F1 on clean docs

**Clopper-Pearson**
$\hat{p} = \text{Beta}^{-1}(1-\alpha;\, s+1,\, n-s)$

**Adaptive attacker (PAIR)**
max over δ ∈ Δ, |δ| ≤ attack_budget → $\hat{p}^{adp}$

**Forging bound**
q / |$K_\tau$|  (log-space)

**χ²-DRO**
$g_\rho(p) = p + \sqrt{(2\rho\; p(1-p))}$  (else 1)

**Combined risk**
$\hat{R}_\tau(\rho) = \min\{1,\; q/|K_\tau| + \max(g_\rho(\hat{p}),\, \hat{p}^{adp})\}$

**Sion game, level 1**
$V_1 = \min_\pi \max_{\text{attack}}$ (SOCP, cvxpy/Clarabel) → π*

**Choice rule**
$\tau^* = \text{argmin}\, \hat{R}_\tau$ s.t. $U_\tau \geq U_{\text{none}} - \eta$

**Significance**
McNemar (paired) · bootstrap CI

**Hand-off to Stage 2**
π* · |K| = min |$K_\tau$| · $\hat{R}_{\tau^*} = \Sigma\, \pi^*_\tau \cdot \hat{R}_\tau$
results/stage1/pi_star.json

## 3 · Direct injection / jailbreak layer (off-the-shelf, no training)

**Output guard**
Llama Guard 3 (1B) · harm categories

**User prompt u**
trusted channel · role owner / member / guest

**Canary check**
secret token leaked ⇒ block

**Input guard**
LlamaFirewall + Prompt Guard 2 (86M)

**Answer to the user**

**Role rules**
policy.yaml: what each role may call

## 4 · Stage 2 — math-hardened CaMeL (online, every request)

**Tool outputs**
emails · files · web (untrusted)

**Privileged LLM**
sees ONLY u → writes a Python-like plan

**CaMeL interpreter (STRICT)**
google-research/camel, vendored unmodified
tracks data + control dependencies

untrusted x

π*, |K|

**Quarantined LLM × m**
run ε: own $\tau_\varepsilon$ = π*, own fresh $x_\varepsilon$, no tools

RQ

**Typed parser**
Pydantic schema · strict / loose string caps

**QIF leakage**
Σ $\log_2|S|$ bits ≤ leakage_cap_bits

**Majority vote**
> m/2 identical → value, else refuse

**Agreement score**
s(x) = (# agreeing runs) / m

**Hoeffding**
P(vote fooled) ≤ $\exp(-2m(\frac{1}{2} - \hat{R}_Q)^2)$

**Capability labels**
SYSTEM ⊑ USER ⊑ UNTRUSTED
$\text{cap}(f(v_1,..)) = \sqcup\, \text{cap}(v_i)$

**Policy table**
config/policy.yaml · strict / confirm_untrusted

**Noninterference (Z3)**
∀ args: UNTRUSTED ⇒ ¬ALLOW · proven at startup

**Conformal risk control**
ask iff s(x) ≥ λ̂, else DENY
$\hat{\lambda} = \min \lambda : (n/(n+1))\, R(\lambda) + 1/(n+1) \leq \alpha_{\text{user}}$

**DENY**

**ALLOW**

**Ask the user**
simulated approve / you in the shell

cap

G, C, F per attack

approved

G, C, F per attack

**Tool call**
AgentDojo environment

G, C, F per attack

## 5 · End-to-end Horizon risk

**Per-attack loss**
$\ell = G + a\,C + b\,F$
G goal hit · C confirm asked · F false refusal

**Loss bound**
$\bar{\ell}$ = Clopper-Pearson upper on mean loss

**Sion game, level 2 (LP)**
$\min_{\pi \in \Pi^{\text{adp}}} \max_w \Sigma\, w\, \bar{\ell}$ · scipy linprog + duals → π*, w*
Π$^{\text{adp}}$ configs with leakage ≤ cap and U ≥ $U_{min}$

**Horizon value**
$V_H = \min\{1,\; \Sigma\pi\, q/|K| + \max(E[L] + \sqrt{(2\rho\ \text{Var}\ L)},\; \Sigma\pi\, \hat{p}^{adp})\}$
(variance form pending author confirmation)

## 0 · Software & infrastructure

**uv + Python 3.11**
pinned uv.lock · scripts/setup.sh

**colibri engine**
Qwen3.6-35B-A3B (4)
writes the report

**Math libraries**
cvxpy + Clarabel (SOCP) · scipy (Beta⁻¹, linprog)
statsmodels · numpy · Z3

**Ollama → Qwen3-8B**
RTX 4060 · 16k ctx · q8 KV · flash attn
P-LLM, Q-LLM, attacker

**MLflow**
results/mlflow.db

**Role proxy**
developer→system · thinking off

**LiteLLM**
one client · on-disk cache

## 6 · Evaluation (§11 runners, logged to MLflow)

**Unit tests (§9)**
pytest · slide values only in tests/

**Smoke test**
does every layer fire?

**CaMeL baseline**
reference code on AgentDojo

**stage1_runner**
static · clean · adaptive · analyze

**agentdojo_runner**
ablations none · stage1 · stage2 · stage1+2 · full
calibrate λ̂ · adaptive attacks

**jailbreak_runner**
detection rate / FPR on every direct set

**Metrics**
ASR = s/n with CP interval · utility · FPR

**PyRIT**
PAIR · TAP · Crescendo
success if judge score ≥ threshold

**garak**
probe suite via guarded_server.py (REST)

## 7 · Outputs & how you use it

**./horizon.sh**
menu · Stage 1 / Stage 2 / jailbreak demos
help · glossary · session logs

**./run_all.sh**
quick · pilot · full · --report-only

**Figures F1–F18**
matplotlib · PDF + PNG + CSV · formula box

**Pipeline diagrams**
Graphviz → docs/

**Horizon.ipynb**
every step, executed (nbclient)

**report/report.pdf**
colibri model analyses figures + CSVs
numbers checked → LaTeX (xelatex)